Retirement and Portfolio Choices: Evidence from Italian Households

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Abstract

This paper examines how retirement affects household portfolio allocation using rich panel data from the Italian Survey on Household Income and Wealth (SHIW) from 2012 to 2022. Exploiting exogenous variation in pension eligibility rules, we estimate a 2SLS with individual and time fixed effects to identify the causal impact of retirement on stock holdings. We find that when the head retires, households increase the share of financial wealth invested in stocks by more than 7 percentage points. This result contrasts sharply with standard portfolio choice theory, which predicts a decreasing equity share with age, but aligns with recent evidence of a negative relationship between background risk and households' stock market participation. When people retire, they are no longer exposed to employment risk. Therefore, they can afford to "replace" their background risk with financial risk if they are not close to the liquidity constraint. This effect is driven by wealthy retirees who had stable careers without unemployment spells. In contrast, poor liquidity-constrained retirees not only invest less in financial markets, but also reduce consumption upon retirement.

Keywords: Stockholdings, Early Retirement, Background Risk
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1 Introduction

Retirement is a turning point in a person's life, as it entails not only changes in daily routines but also a substantial reorganization of financial resources. Understanding how investment decisions change at retirement matters from a policy perspective, since households' wealth reallocation at this life-cycle stage can influence asset demand and, in turn, asset returns (Auclert et al., 2021). This is even more salient in a country like Italy, where the share of retirees relative to the working population is expected to increase in the coming decades (INPS, 2025), making retirement-related portfolio and consumption decisions increasingly relevant for aggregate saving.

Yet what retirement means for households' stock holdings remains somewhat unclear. On the one hand, standard life cycle models (Cocco et al. 2005) suggest that stock market exposure should decrease with age. The key reason is that human capital and stock market returns are independent and therefore complementary, as both help minimize background risk. Therefore, the share of wealth invested in stocks should decrease upon retirement: when people retire, they are no longer exposed to employment risk, and so they should reduce their exposure to the stock market. However, as noted by Catherine (2021) and Catherine et al. (2024), standard life cycle models of portfolio choice with uninsurable labor income risk fail to match observed stock market participation profiles, at least without assuming an unreasonably high level of risk aversion. Moreover, empirical evidence (Fagereng et al. 2018, Guiso et al. 1996) shows a negative relation between uninsurable income risk and stock market participation. These papers suggest that households should reduce their stock holdings when they retire. The intuition is that, because they are no longer exposed to employment risk and their background risk falls significantly, they can afford to replace income risk with financial risk.

In this paper, we empirically investigate the impact of retirement on stock holdings. To address the endogeneity of the retirement decision, we implement a two-stage least squares (2SLS) analysis with individual and time fixed effects (TWFEs) in which pension eligibility - determined by the retirement schemes available in Italy between 2012 and 2022 - is used as an instrument for actual retirement. This approach allows us to compare the same individual while working and after retirement, controlling for common time shocks and individual-specific unobserved heterogeneity that could affect portfolio choices over time. We find that when the head retires, households increase the share of financial wealth invested in stocks by 7.5 percentage points. However, participation in the stock market is unaffected. This result is consistent with the idea that when households retire, they are no longer exposed to employment risk. This means that their overall background risk decreases and they can "afford" to increase their exposure to financial risk. We also find a negative effect of retirement on consumption. Upon retirement, nondurable consumption drops by 23%, a much larger effect than Battistin et al. (2009) finds. This reduction in consumption upon retirement may point to the well-known retirement consumption puzzle. Given that retirement is a predictable event that entails an expected drop in income, people should smooth consumption accordingly and not cut consump-

tion upon retirement. However, we argue that this is only apparently a puzzle, as this effect is driven entirely by liquidity-constrained retirees, who are therefore sensitive to expected income changes.

We draw data from the Bank of Italy's *Survey of Household, Income, and Wealth* (SHIW). This biannual survey[1] collects detailed information on the wealth composition of a representative panel of Italian households. The survey also asks respondents to report their years of contribution - i.e., the number of years they have worked under a regular contract - which allows us to determine whether and when people are eligible for retirement. More importantly, SHIW has an important panel we use to control for time-invariant heterogeneity in portfolio returns (Fagereng et al. 2020).

Between 2012 and 2022, Italian workers had access to three main retirement pathways. The first, which we will call *seniority*, allowed individuals to retire after accumulating 42 years of contributions for men and 41 for women. The second option, which we will call *old age*, allows retirement at age 67 with at least 20 years of contributions, for both men and women. Finally, between 2019 and 2022, individuals could retire with 38 years of contributions and a minimum age of 62 (raised to 64 in 2022); we call this option *early retirement*.

We use these eligibility rules to construct our instrumental variable. Specifically, we define an individual as eligible for retirement if, in a given year, she meets the requirements for at least one of the retirement schemes described above. Our identification strategy relies on the fact that institutional rules determine eligibility and are thus exogenous to individual portfolio decisions, while strongly predicting the probability of retiring once we condition on time-invariant unobserved heterogeneity. Indeed, a potential concern is that eligibility rules are partly determined by accumulated years of contributions, which, in turn, reflect individuals' career histories. Because career stability and earnings trajectories may correlate with portfolio decisions, this could generate endogeneity in our setting. In our 2SLS estimation, we address this issue in two ways: first, we use a TWFE strategy, which lets us compare the same individual across working and retirement, thereby absorbing time-invariant unobserved heterogeneity related to career choices.

Our identification strategy rests on the fact that workers do not choose eligibility requirements, nor can they self-select into them. However, individuals have different working histories and careers, which reflect both exogenous shocks in their lives and choices driven by heterogeneous preferences. We address this with a large battery of controls, including information on past unemployment spells.

Stock holdings and consumption may still follow different trends years before people become eligible for retirement. With a staggered event study, we show that people had similar trends up to 10 years before they became eligible for retirement. We show this also holds for incomes other than working income (which, of course, goes to 0 due to retirement). Before eligibility, trends of real estate income (i.e., rents) and financial income (i.e., coupons and dividends, on top of realized capital gains) are now significantly different. However, we do not control for

[1] The SHIW is typically conducted every two years, except the 2016 and 2020 waves.

them because, after retirement, people change their investment strategies, and their investment income also changes. Therefore, they would arguably be bad controls.

We then investigate which category of retirees is actually investing more in stocks. To do so, we run a reduced-form analysis in which we directly regress the outcome variables on the eligibility dummies. We find that the increase in stock holdings is almost entirely driven by "seniority" retirees, while it's indistinguishable from 0 for *old age* and *early retirees*. In contrast, only *old age* retirees cut consumption upon retirement, while the other retirees behave according to standard theory and smooth consumption upon retirement. The common element connecting these two results is that *old age* retirees are poorer than the other retiree categories. They experienced more discontinuous and precarious careers, characterized by more frequent and longer unemployment spells. As a consequence, they had less opportunity to accumulate wealth and exited the labor market with a lower severance payment. For this reason, they are liquidity constrained and have to cut consumption when they stop receiving labor income. In this sense, we do argue that this is not really a consumption puzzle, as the drop in consumption upon retirement happens for those people who, according to theory, are sensitive to expected income fluctuations (Jappelli and Pistaferri 2017). In contrast, *seniority* retirees had a more stable career path and exited their job with a higher severance payment. Therefore, they can smooth consumption upon retirement and invest their severance payment in stocks. Comparing *old age* with seniority retirees, we document that the former have spent one year in unemployment on average, are less likely to be homeowners, and have a substantially lower amount of cash and net wealth.

Related Literature. Our paper is connected to different strands of literature and contributes to them along several dimensions. First, it mirrors the literature on the *retirement consumption puzzle* (see Banks et al., 1998). Recent empirical work by Kolsrud et al. (2024) and Olafsson and Pagel (2024) in Scandinavian countries has shown an economically and statistically significant decline in consumption upon retirement. Results like these, also observed in Italy by Battistin et al. (2009), run counter to standard economic theory, which assumes consumers do not alter consumption in response to predicted income changes. Aguiar and Hurst (2005) rationalizes this puzzle by emphasizing the difference between consumption and expenditure, as the former also includes home production and the depreciation of durables (like TVs, laptops, cars, etc.), which survey data measure less precisely than expenditure. Another explanation for the puzzle is the complementarity between labor and specific consumption items. Commuting costs are a perfect example: as people retire, they no longer need to commute to work, and thus, transportation expenditure decreases. We speak to this literature by first shifting the focus from consumption to investment behavior. Moreover, we provide evidence that the drop in consumption concerns only old-age retirees, who, for the reasons outlined above, are more likely to be liquidity constrained. By contrast, we find no convincing evidence of a drop in consumption for senior and early retirees, who are, on average, further from borrowing constraints and more likely to behave as Permanent Income Hypothesis (PIH) maximizers. In this sense, there is no "retirement consumption puzzle", since it reflects the effect of liquidity

constraints on poor retirees.

Second, we contribute to the extensive literature in consumption and household finance on the impact of income risk on consumption and portfolio choice. The papers in this strand, either by estimating structural models or relying on natural experiments, test the implications of the precautionary saving model and the buffer stock model[2]. As mentioned in the first few lines, retirement also reduces income-risk exposure; therefore, our results may reflect the reduced-form impact of background risk on portfolio choice. In this regard, our paper is closely related to Fagereng et al. (2018), which examines the impact of uninsured wage risk on participation in the stock market in Norway. Using employee-employer matched and tax record data, and applying the methodology of Guiso et al. (2005) to extract the component of wage volatility not insured by the employer, they show that risk reduces stock holdings, and that the effect is greater for households belonging to the bottom of the wealth distribution. Guiso et al. (1996) find, in the 1989 SHIW cross-section, a negative correlation between risky-asset holdings and the subjective variance of next-period income growth, elicited through survey questions. Angerer and Lam (2009) show that portfolio choices are mostly affected by permanent income volatility, but not by transitory income risk. Silva et al. (2025) show that going public (i.e., employers' IPOs) reduces wage volatility and increases holdings of risky assets. We differ from these papers in that we focus on variation in risk exposure due to retirement; in other words, we compare the same individuals before and after retirement, isolating the effect of income risk reduction within the life cycle.

Third, we contribute to the literature on households' participation in the stock market and how this changes in response to wealth shocks. Stock market participation among households is well known to be low and mostly concentrated among the top quantiles of the wealth distribution (see Fagereng et al. 2017 and Fagereng et al. 2020 in Norway). Van Rooij et al., 2011 point to low financial literacy (measured by "The Big Three" as in Lusardi, 2008) as the main reason in the United States. Moreover, Lusardi et al. (2017) shows that differences in financial literacy are among the main drivers of cross-sectional wealth inequality in the United States. More recently, Catherine (2021) argues that the low participation in the stock market of the bottom of the wealth distribution is due to the cyclicality of the skewness of earnings shocks, which tends to be negative (more mass on the left tail) in bad times (i.e., recessions, usually corresponding to stock market crashes). This mechanism is confirmed by Catherine et al. (2024), who use Swedish data and show that the (left) skewness of earnings shocks is negatively correlated with the share of equity holdings. Cioffi (2021) argues that people along the wealth distribution are exposed heterogeneously to shocks in asset returns. Therefore, as they become wealthier, they move away from safe assets and invest first in housing, then in the stock market.

Christelis et al. (2024) estimates the causal effect of wealth shocks on stock holdings by randomly assigning hypothetical lottery wins to respondents to the Consumer Expectation Survey (CES). They find that the increase in stock market participation, though statistically

[2] See Jappelli and Pistaferri (2017) for a review of the main theoretical and empirical contribution and **?** for a recent test of the buffer-stock model by Carroll and Samwick (1997) and Deaton (1991).

significant, is contained (around 8.4 percentage points), especially for lottery gains as high as €50.000. Also, the share of wealth invested in stocks (the intensive margin) increases only slightly (by 1.7 percentage points for a hypothetical wealth shock of €50.000). This result indicates that the assumption of Constant Relative Risk Aversion (CRRA) preferences is a reasonable one for most investors, as heterogeneous stock market participation is mostly explained by risk aversion heterogeneity[3]. We contribute to this literature first by showing that retirement increases stock market participation in Italy and then that the estimated causal effect of retirement on stock market participation remains stable even when controlling for household wealth, suggesting that our results are not driven by mechanical wealth changes, but rather by shifts in risk exposure associated with the transition to retirement.

Deeply connected to our paper are also the works that study the impact of pension reforms in the 1990s implemented in Italy on retirement expectations (see Bottazzi et al., 2006), savings and portfolio composition (see, for instance, Attanasio and Brugiavini, 2003, Bottazzi et al., 2011, and Daminato and Padula, 2024). Unlike these studies, however, we do not look at the impact of pension reforms affecting people who are currently working, but at how stock holdings change in the transition from work to retirement.

Our paper is close to Castaldo (2024). Following an RDD approach as in Battistin et al. (2009), he examines the impact of an infusion of liquidity upon retirement (i.e., the severance payment) on bond holdings, using SHIW data from 1993 to 2016. He finds an 8-percentage-point increase in stock market participation. We augment his analysis on three dimensions. First, using a different time window and an additional retirement option (i.e., Q100), we also show an effect on the intensive margin, which we define as the share of financial wealth allocated to stocks. Second, we go deeper into the mechanisms, showing that the effect is significant only for retirees with strong career paths and thus high severance payments. This result mirrors the result that consumption drops only for old-age retirees, who are, for the same reasons, more likely to be liquidity constrained. Third, we also exploit the panel component to isolate the effect of interest from unobserved heterogeneity in individual returns to wealth, which arguably affect both eligibility for retirement (because they might affect labor supply elasticities and the taste/distaste for work) and portfolio decisions over the life cycle (Fagereng et al. 2020). The panel component also lets us observe the dynamics of a person's stock holdings before and after retirement in an event-study style. In addition, we also believe that the time window matters for identification. We deliberately exclude the 2008-2012 crisis from our analysis because of the Fornero reform (2011)[4]. The reform aimed to restore the sustainability of the National pension system amid a financial crisis that arguably affected households' stock-holding decisions in a non-homogeneous way, and thus not captured by time FEs. Therefore, time dummies alone are not enough to purge heterogeneous households' exposure to the crisis. Our sample, by contrast, includes the COVID-19 period. A potential concern is that the pandemic may have affected portfolio choices differently according to households' pension eligibility. However, the

[3] In the standard portfolio choice as in Merton (1969), the share of wealth invested in risky assets is not a function of wealth itself. It is a decreasing function of risk aversion and the variance of the asset return.

[4] The reform was implemented in 2011 and was effective starting January 1st 2012

pandemic did not directly change the pension eligibility rules exploited by our instrumental variable. Hence, the main concern here is not a mechanical change in the instrument per se, but heterogeneous portfolio responses during the pandemic.

Finally, we also address the literature on the macro-finance consequences of population aging. Auclert et al. (2021) argues that the aging in the population will increase the wealth-to-income ratio, and thus lower returns on wealth. Our results suggest that a larger fraction of the population will also increase their holdings on the intensive margin. Therefore, stock market returns will decline as higher retail demand pushes prices up.

Outline. The paper is structured as follows. Section 2 describes the data, the retirement options we use to build the instrument, and the sample selection. Section 3 depicts the empirical strategy and the baseline results. We also discuss the instrument's validity, while Section 4 shows the main mechanisms that rationalize our results and some heterogeneity analysis. Finally, Section 5 concludes and describes our next steps.

2 Data and Retirement Options

Sample Construction and Retirement Options. We draw on the Bank of Italy's Survey of Household Income and Wealth (SHIW), a comprehensive dataset that provides insights into the financial lives of Italian households. SHIW contains detailed (self-reported) information on demographics, income, consumption, and wealth. In particular, respondents are asked to self-report in detail their holdings of stocks (both public and private), bonds, bank accounts, and other forms of financial investments. They are also asked to report their liabilities and the value of the real estate properties they own. Moreover, SHIW also includes information on individuals' years of contribution, i.e., how many years respondents have worked under a regular contract, which is crucial for constructing our instrumental variable. The convenient feature of SHIW is its panel component: each wave includes around 50% of households interviewed in the previous wave. This allows us to track individuals over time and compare the same individuals before and after retirement. This lets us control for time-invariant unobserved heterogeneity in risk aversion and portfolio returns (Fagereng et al. 2020).

We select the sample following Jappelli and Pistaferri (2020, 2025). To start, we keep only household heads[5] who were employed in 2012. We define an individual as retired if she is not working and receives a retirement pension[6]. To ensure we track the same individuals over time, we drop observations with inconsistent demographics. Therefore, we drop households whose head changes sex over time and with impossible ages[7]. For instance, if a person is

[5] Restricting the analysis to heads prevents complications arising from changes in household composition or from the identity of the head switching across survey waves.

[6] Our dataset contains detailed information about the nature of the pension income that can come from several sources, including old-age, seniority, disability, and survivor pensions. In this analysis, we exclude individuals receiving disability or survivor pensions and retain only those receiving old-age or seniority pensions, as these are directly linked to retirement decisions and eligibility.

[7] These cases likely reflect changes in the identity of the household head across waves or reporting errors, rather than actual demographic variation. We exclude them to ensure that the same individual is consistently

30 years old in 2012, she must be 32 in 2014, 34 in 2016, 38 in 2020, and 40 in 2022. To minimize concerns about measurement error in (self-reported) years of contribution, we keep only those individuals for whom the increase in years of contribution is less than or equal to the time difference between consecutive waves. We drop individuals who report fewer years of contribution than the preceding wave. After applying these sample selection criteria, we are left with around 1300 observations, pooled from the waves of 2012, 2014, 2016, 2020, and 2022.

In the period under analysis, people could leave the labor market following three different paths to retirement. Under the first path, which we call *seniority*, workers might retire, regardless of age, as soon as they accumulate 42 years of contributions if they are men and 41 if they are women. The second path, which we call *old age*, requires workers, both men and women, to be at least 67 years old and to have 20 years of contributions. Finally, between 2019 and 2021, a third option known as *Quota 100* (Q100) was introduced. Under Q100, people aged 62 with 38 years of contributions could choose to retire. In 2022, Q100 was replaced by *Quota 102* (Q102), which required age 64 and 38 years of contributions. We refer to these two last options together as *early retirement*, as they generally represented a milder eligibility criterion, at the cost of lower pension wealth upon retirement.

People could choose which option to use and, typically, their choice is dictated by their careers and their earnings trajectories during their working history. For example, seniority represents the typical option used to retire. It offers the largest pension benefit[8]. This is because, if a person has a continuous working life, not characterized by long unemployment spells, she reaches the eligibility criteria for seniority in her early 60s, even earlier if she started working in her teenage years. By contrast, old age entails a much lower pension wealth, and it is typically "chosen" by individuals who have experienced large unemployment spells during their lives. This may reflect that irregular career paths often prevent individuals from accumulating the required contribution years for more generous schemes. In the Italian public pension system, pension benefits are commonly calculated based on both contribution years and past earnings, using a contribution-based or mixed formula depending on the year of entry into the labor market. Under the contribution-based regime, which applies to workers who started contributing after 1996, the benefit equals the sum of all contributions paid over the working life, capitalized at an annual rate linked to GDP growth, multiplied by a conversion coefficient that depends on the retirement age. Workers with earlier contribution histories fall under a mixed regime that combines earnings-related and contribution-based components, resulting in higher replacement rates.

We use age and years of contribution to define eligibility for the three retirement options mentioned above: seniority, old age, and early retirement. To define eligibility, we assume the following assignment rule (or preference order) among the three schemes: (1) seniority, (2) early, and (3) old age. If a person satisfies the requirements for more than one option in a given year, we assign her to the highest-ranking scheme for which she is eligible. This rule is

tracked over time.

[8] According to INPS, 2025, in 2024, roughly 30% of new pensions in Italy were seniority/early retirement pensions, confirming the widespread relevance of this option among workers with continuous careers.

not meant to reflect individual preferences, but rather the institutional structure and observed behavior in the data. In practice, workers tend to prioritize seniority retirement whenever possible, as it provides a more generous pension benefit and does not restrict post-retirement employment, unlike early retirement. By contrast, individuals typically access old-age pensions when they cannot meet the contribution requirements for the other schemes. This institutional background helps explain the observed take-up patterns: the take-up rate for early retirement is lower than for seniority. At the same time, old-age pensions represent a residual option for workers with fragmented careers[9]. In particular, we define:

$$
\begin{aligned}
Seniority_{it} &= \begin{cases} 1 \quad \text{if} & yc_{it} \geq 42 \text{ and } female_i = 0 \\ 1 \quad \text{if} & yc_{it} \geq 41 \text{ and } female_i = 1 \\ 0 \quad \text{otherwise} & \end{cases} \\
Old\,age_{it} &= \begin{cases} 1 \quad \text{if} & age_{it} \geq 67 \text{ and } yc_{it} \geq 20 \\ 0 \quad \text{otherwise} & \end{cases} \\
Early_{it} &= \begin{cases} 1 \quad \text{if} & yc_{it} \geq 38 \text{ and } age_{it} \geq 62 \text{ and } t = 2020 \\ 1 \quad \text{if} & yc_{it} \geq 38 \text{ and } age_{it} \geq 64 \text{ and } t = 2022 \\ 0 \quad \text{otherwise} & \end{cases}
\end{aligned}
\tag{1}
$$

where yc_{it} is the years worked by individuals i at time t. Following the eligibility rules defined above, we construct our instrumental variable for actual retirement as follows:

$$
Eligible_{it} = \begin{cases} 1 \quad \text{if} & seniority_{it} = 1 \\ 1 \quad \text{if} & old\,age_{it} = 1,\ seniority_{it} = 0,\ ealry_{it} = 0 \\ 1 \quad \text{if} & early_{it} = 1,\ seniority_{it} = 0 \\ 0 \quad \text{otherwise} & \end{cases}
\tag{2}
$$

In this setting, we implement a 2SLS regression in which we instrument retirement status with eligibility, as defined above, controlling for both individual and time shocks. Our identification strategy relies on the assumption that, conditional on individual and time fixed effects, eligibility for retirement affects portfolio decisions only through actual retirement.

We use an instrumental variable because the retirement decision may be endogenous to portfolio composition. Retirement status may be correlated with unobserved factors that also influence portfolio choices, such as unmeasured expectations about future income, health shocks, or individual preferences toward risk and saving. For instance, risk-tolerant workers might both retire earlier and invest more in stocks, leading to an upward bias in OLS estimates. Using pension eligibility as an instrument isolates the exogenous variation in retirement induced by

[9] As a robustness check in Appendix A, we perform the main analysis by dropping this assumption and allowing individuals to be contemporaneously eligible for more options. Results are practically unchanged. The main reason is that there are very few people in the sample eligible for more than one option at the same time.

institutional rules, allowing us to recover the causal effect of retirement on portfolio allocation.

Descriptive Evidence. Figure 2 shows the take-up ratios, i.e., the ratio of eligible individuals to actual retirees, for the three retirement options. As we can see, almost 88% of old age-eligible individuals are retired, while the take-up ratio goes down to slightly above 80% for seniority. The share of early-eligible individuals who are retired is below 80%.

The main message of Figure 1 is that most people eligible for retirement choose to retire, consistent with Battistin et al. (2009).

Figure 2 displays the relationship between actual retirement status and the distance to retirement eligibility. The Figure shows an overall positive relationship between distance to retirement eligibility and actual retirement. In particular, the share of retirement-eligible individuals who are in retirement is 100% six years later. This pattern confirms the predictive power of eligibility for actual retirement behavior and validates the relevance of our instrument. Figure 2 also supports the plausibility of the monotonicity assumption: it is unlikely that anyone in the sample would have retired if they had not been eligible, while not retiring after becoming eligible remains rare.

In Table 1, we report the descriptive statistics for non-eligible, early retirement-age eligible, seniority-eligible, and old-age eligible. As the first column shows, 4% of the non-eligible subsample is retired. This is because people sometimes benefit from early retirement options offered during mass layoffs or business restructuring.

Seniority-eligible individuals hold more cash on hand (income + financial assets) than old-age- and early-eligible individuals, and fewer declare long unemployment spells (more than 12 consecutive months). Moreover, seniority-eligible individuals hold a higher share of stocks and are more likely to participate in the financial market than other retirement-eligible individuals.

In Appendix A we provide additional descriptive evidence that complements our analysis. In particular, Figure A.5 shows the cross-sectional distribution of overall financial wealth (top-left), liabilities (top-right), cash (bottom-left), and stocks (bottom-right). While a substantial part of the sample has little overall financial wealth (around 40%), a sizable portion (around 10%) also holds €40000 in stocks. Figures A.7 and A.8 show, on the other hand, the pooled age profiles of financial wealth and stocks, both on the intensive and the extensive margin. As is standard in the literature, these profiles are hump-shaped: they rise until retirement (approximately age 60) and then decline. Finally, Figure A.8 shows the composition of financial wealth by deciles of the wealth distribution. While the bottom half of the distribution holds cash primarily, the top 20% holds non-negligible amounts of bonds, stocks, and mutual funds.

3 Empirical Strategy and Baseline Results

Our empirical strategy uses 2SLS, instrumenting actual retirement with retirement eligibility as defined above. Our main equation is:

$$Y_{it} = \alpha_i + \lambda_t + \beta Retired_{it} + \varphi \mathbf{X}_{it} + \varepsilon_{it} \tag{3}$$

In (3), y_{it} is either the share of stocks held in i's portfolio at time t as a share of gross financial wealth, or a dummy equal to 1 if i holds stocks at time t. Both variables are multiplied by 100. The control vector $\mathbf{X}_{it}$ includes age (to control for the life cycle), education, family size, civil status, and, in the most demanding specification, net wealth, i.e., the sum of financial assets, liabilities, and real estate assets. The variable $Retired_{it}$ is a dummy equal to 1 if i is retired at time t. α_i and λ_t are individual and time fixed effects. The former are intended to purge the analysis of time-invariant characteristics that affect retirement and stock holdings simultaneously. One example is whether an individual suffered financial losses in the past that scared her and forced her to postpone retirement. The latter, on the other hand, captures common shocks that affect individuals in a presumably homogeneous way, such as bad news about the sustainability of the National Social Security Institute. Fixed effects also control for heterogeneity in returns to wealth, as documented by Fagereng et al. (2020). Finally, ε_{it} is the error term clustered at the individual level, to account for serial correlation in unobserved shocks over time for each individual.

Threats to Identification. Estimating (3) by OLS is going to lead to an inconsistent estimate of β because people choose when (and how) to retire, and this choice is likely to be correlated with time-variant unobservable characteristics affecting portfolio decisions as well. One potential source of endogeneity arises from unobservable preferences for work. Individuals who enjoy working or derive non-monetary satisfaction from their jobs are likely to postpone retirement. At the same time, because they work longer, they accumulate higher lifetime wealth, which may translate into a greater ability to invest in riskier assets. Then, the observed association between retirement and portfolio reallocation would partly reflect differences in lifetime earnings driven by unobserved preferences rather than the causal effect of retirement itself.

Another problem is reverse causality: an individual might earn very high returns from the financial market, which convinces her to stop working. Georgarakos et al. (2025), randomly providing individuals with hypothetical lottery gains, show that people respond by reducing labor supply on both the intensive and extensive margins. Cesarini et al. (2017), Fagereng et al. (2021), and Golosov et al. (2024), in Sweden, Norway, and the United States, respectively, show that lottery winners persistently reduce their gross and net earnings.

To tackle these issues, we instrument actual retirement with eligibility. Pension eligibility is determined exclusively by institutional rules based on age and contribution history, not on individual preferences, expectations, or financial behavior. Conditional on individual and time-fixed effects, these institutional thresholds are exogenous to unobservable factors such as a taste for work or expectations about future income risk, as well as to shocks to financial returns that might directly influence portfolio choices. In other words, eligibility affects portfolio decisions only through its impact on the probability of retiring, satisfying the exclusion restriction. The first stage we are going to estimate is the following:

$$Retired_{it} = \alpha_i + \lambda_t + \gamma\, Eligible_{it} + \rho \mathbf{X}_{it} + v_{it} \tag{4}$$

In (4), $Eligible_{it}$ is a dummy variable defined as above. Our coefficient of interest is β, the LATE of retirement on the portfolio, i.e., the effect on compliers. In this context, compliers are those individuals who retired *because* they became eligible, so they satisfy the age and years-of-contribution requirements for at least one of the three retirement options.

Results on Stock holdings. Table 2 shows the baseline results for all stocks, both on the intensive (first three columns) and the extensive (last three columns) margins. The LATE of retirement on the share of stocks is around 7 percentage points, with no effect on participation in the stock market. Adding demographic controls makes little difference, as neither the point estimates nor the standard errors change significantly. Nor does controlling for wealth change anything: the coefficient attached to wealth is both imprecisely estimated and economically insignificant (a €1000 increase in net wealth increases the share of stocks by 0.01 percentage points). This result is consistent with Christelis et al. (2024), suggesting that heterogeneity in portfolio composition is driven mainly by differences in risk aversion rather than wealth levels. Table 3 and Table 4 decompose the result for public stocks - i.e., stocks traded in the stock market - and private stocks. The effect is present only for the former and only for the intensive margin.

The Effect on Consumption. Using Italian data from 1989 to 2004 Battistin et al. (2009) show that households reduce consumption when they retire. Recent papers use administrative and transaction-level data in Scandinavian countries (see Kolsrud et al. 2024 and Olafsson and Pagel 2024). This evidence sharply contrasts with the permanent income hypothesis (Hall 1978, Friedman 1957), which posits that households should vary consumption only in response to unexpected changes in permanent income. Since retirement is a reasonably expected event in people's lives, and people can form expectations about their pension income, consumption should not change. In this section, we look at the impact of retirement on consumption in the same way as before. In particular, we estimate the following equation:

$$\log C_{it} = \alpha_i + \lambda_t + \beta \, Retired_{it} + \varphi \mathbf{X}_{it} + \varepsilon_{it} \tag{5}$$

In equation (5), the dependent variable is the log of nondurable consumption, multiplied by 100, while the first stage is the same as before. Unlike Battistin et al. (2009), we include individual and time fixed effects in the specification. On the other hand, (5) resembles the specification used by Olafsson and Pagel (2024), except that we estimate it with 2SLS, instrumenting retirement with eligibility. Of course, our data have the caveat that we use self-reported consumption, while they use measurement error-free, transaction-level data. Table 5 shows that nondurable consumption drops by 23% and that the estimate is robust to adding controls. Table 6 shows what happens to other assets. The odd columns show the intensive margin, while the even columns show the extensive margin. First, we notice a decline in cash holdings, though it's not statistically significant.

Assessing the Validity of the Instrument. For the instrument to really estimate the

LATE of retirement on stock holdings, it has to satisfy two requirements. The first is relevance: we need retirement eligibility to predict actual retirement. The second requirement, the exclusion restriction, is that the instrument affects stock holdings only through its effect on retirement. If the instrument were not relevant, we would expect no strong statistical relationship between eligibility and actual retirement - that is, becoming eligible would not significantly predict the probability of retiring. To test this, we look at the statistical relationship between eligibility and actual retirement. Then, we estimate the first stage in equation (4) by OLS and report the results in Table 6. The instrument is strongly correlated with actual retirement: being eligible for retirement makes actual retirement more likely by 60 percentage points. Controlling for individual and time FEs, as well as for the same variables we control for in equation (3), changes the coefficient but not the strength of the first stage. The F-stat of the excluded instrument is always above 60, well above the benchmark recommended in the literature for instrument relevance.

The second requirement for a valid instrument is the exclusion restriction: the eligibility must affect shock holdings only through retirement. As is well understood, this assumption is untestable. Nevertheless, we can make arguments in its defense. Our first argument is that pension eligibility depends on variables that are difficult to manipulate: age and years of contribution. However, individuals have different working histories and careers, which reflect both exogenous shocks in their lives and choices shaped by heterogeneous preferences. Unfortunately, our panel does not go deep enough to observe household heads' complete working histories. Some of these unobservable characteristics might be time-varying, and so not captured by individual fixed effects. If time-varying unobserved heterogeneity influences consumption and stock-holding decisions, our 2SLS violates the exclusion restrictions. We address this first with a large set of controls. In addition to demographic characteristics such as civil status, education, and family size, we also control for how many years people have spent unemployed in their lives. While this is not the best control for past work history, it is still an important piece of information we can leverage. Indeed, Bertheau et al. (2023), using administrative data on several OECD countries, show that Italy, in the OECD, is among the countries with the highest job displacement costs: after unemployment spells, even when people find a new job, earnings are significantly and persistently below the level before the unemployment spell occurred. In other words, losing your job in Italy, and a fortiori being unemployed for a long time, has long-run effects on people's income and, on average, wealth. We also control for real estate wealth and whether households are homeowners. Although we look at financial wealth, households might choose different investment strategies depending on whether they own the home they live in. When we control for real estate wealth, results hardly move.

We then check for pre-trends with respect to eligibility. We want to test whether stock holdings differed systematically before people became eligible for retirement. We therefore estimate the following event study equation both for the intensive and the extensive margin of portfolio choice, where D_{-k} and D_k represent the distance, before and after, to retirement

eligibility:

$$Y_{it} = \alpha_i + \lambda_t + \sum_{k=-10}^{-2} \gamma_k D_{ik} + \sum_{k=2}^{6} \gamma_k D_{ik} + \rho \mathbf{X}_{it} + u_{it} \tag{6}$$

Figure 3 shows the estimated coefficients $\hat{\gamma}_{-k}$ and $\hat{\gamma}_k$, alongside 95% confidence intervals.

The Figure plots the evolution of the share of stocks held (left panel) and the probability of participating in the stock market (right panel) as a function of years before and after eligibility (ranging from 10 years before to 6 years after becoming eligible for retirement). The p-value for the test that all pre-trend coefficients are jointly equal to 0 is 0.12 for the intensive margin and 1.61 for the extensive margin. In other words, in the pre-eligibility period, the OLS coefficients are close to zero and statistically insignificant, suggesting no pre-trends in portfolio allocation. Immediately after eligibility, however, there is a sharp increase: the stock share jumps, as does stock market participation. Therefore, we cannot reject the null hypothesis of no effect in the years before individuals become eligible for retirement.

As a final test, we look at returns from financial and real estate assets before actual retirement and before eligibility. The concern is that systematic differences in asset returns, driven by particularly good (or bad) investments, might be the true cause of changes in portfolio composition. For instance, if individuals who retire happen to experience unusually high returns on their financial assets, this could mechanically inflate their stock holdings, confounding our estimates. Figure 4 shows that there is no clear trend before eligibility

4 Reduced Form Analysis and Heterogeneity

Reduced-Form. In this section, we dig deeper into the baseline result and investigate which eligibility category is driving it. To do so, we run a reduced-form analysis, regressing the outcome variable on the three eligibility dummies defined in Section 2. In particular, we estimate the following equation by OLS and report the estimated coefficients in Table 7.

$$Y_{it} = \alpha_i + \lambda_t + \theta Early_{it} + \delta Seniority_{it} + \omega Old\,age_{it} + \kappa \mathbf{X}_{it} + v_{it} \tag{7}$$

The three eligibility dummies are mutually exclusive. If an individual qualifies for more than one option in a given year, we assign her to the highest-ranking scheme according to the preference order defined in Section 2, namely seniority, early retirement, and old-age. This ensures that each person-year observation is classified under one, and only one, eligibility rule.

Looking at the intensive margin (column 1) of Table 8, only the coefficient attached to $seniority_{it}$ is positive and statistically different from zero. In particular, eligibility for seniority retirement is associated with a share of stocks (listed and unlisted) over financial wealth that is higher by around 2 percentage points. In contrast, the coefficients on *early* and *old age* are not statistically different from zero at 95%. Turning to the extensive margin (column 2), we find that the coefficient on *seniority* is statistically different from 0, even though it does not affect the average retirement stock market participation. Finally, in column 3 of the same table, only

old-age retirees reduce consumption upon retirement, while the consumption of *seniority* and *q100* does not change. This is because seniority-eligible retirees have, on average, had a more stable career path. A more stable career path thus translates into a lower realized earnings volatility over the life cycle, a higher level of cash on hand upon retirement, and then a weaker precautionary saving motive. For this reason, when they retire, they are more willing to invest in stocks to earn higher returns. Additionally, their financial situation upon retirement is more predictable, allowing them to treat retirement as a planned, rather than reactive, transition. This forward-looking behavior likely reinforces their willingness to re-optimize their portfolio allocation, taking advantage of their improved ability to bear financial risk. On the other hand, old age-eligible individuals have, on average, had a highly unstable career path, perhaps with irregular contracts and/or long unemployment spells. Therefore, upon retirement, they have much lower cash-on-hand levels and still have a strong precautionary motive for saving, driven by the closeness to the liquidity constraint. As a result, these individuals tend to maintain a more conservative asset allocation, prioritizing liquidity and capital preservation over risky investments. This is the same reason why *old-age* retirees are the only ones to drop consumption after retirement: they are liquidity constrained, and therefore their consumption is sensitive to predictable income changes (Jappelli and Pistaferri 2000 and Jappelli and Pistaferri 2017)

Figure 5 compares *old-age* and seniority retirees across four dimensions: period of unemployment, home ownership, net wealth, and bank account. Old-age retirees have, on average, experienced one more year of unemployment in their lives, are less likely to be homeowners, and hold much less wealth than seniority retirees. This confirms that the former are, on average, much poorer than the latter and therefore less willing to invest in the stock market.

5 Conclusions and Next Steps

In this paper, we estimate the causal effect of retirement on stock holdings. We instrument retirement with eligibility, leveraging the Italian institutional setting for retirement and exploiting the panel dimension of the Bank of Italy SHIW data. The novelty of our approach lies in that we control for unobserved heterogeneity driving portfolio and consumption decisions (Fagereng et al. 2020), differentiating our work from Battistin et al. (2009), Castaldo (2024) and Olafsson and Pagel (2024). We find that when household heads retire, they increase the share of financial wealth invested in stocks by 7.5 percentage points. We argue this happens because, when people retire, they are no longer exposed to employment risk and can replace it with financial risk. This interpretation is consistent with recent papers discussing the negative impact of background risk and its cyclicality on stock market participation. Moreover, we estimate that consumption drops by 23% upon retirement. To assess the instrument's validity, beyond conditioning on an informative set of controls, we show that, before eligibility, stock holdings and consumption were on parallel trends.

Digging deeper into which categories of retirees drive the results, we find that only seniority retirees increase their stock holdings, while only old-age retirees cut their consumption. The

common element connecting these two results is liquidity constraints that bind heterogeneously across retiree categories. Seniority retirees have experienced less time in unemployment, are more likely to be homeowners, and hold much more cash than old-age retirees, who have had more discontinuous and unstable career experiences. For this reason, only the former have the resources to buy stocks, while the latter, being liquidity constrained, must cut consumption as their income falls. Regarding the last result, we show no consumption retirement puzzle: while "normal" retirees smooth consumption around retirement, as predicted by PIH, liquidity-constrained retirees are sensitive to predicted changes in income.

However, an additional channel could explain our results. As people retire, they have more time to read financial news and thus increase their financial literacy. As argued by Jappelli and Padula (2013) and Lusardi et al. (2017), this leads them to invest more in the stock market. Unfortunately, SHIW does not have continuous data on financial literacy, making it impossible to directly understand the role of post-work investment in financial literacy in driving our empirical results. That's why the next step of this paper is to build a life cycle model of portfolio choice, endogenous retirement (as in Daminato and Padula 2024) and endogenous investment in financial literacy (as in Jappelli and Padula 2013). By calibrating the model to the empirical results, we will be able to estimate the strength of the financial literacy channel.

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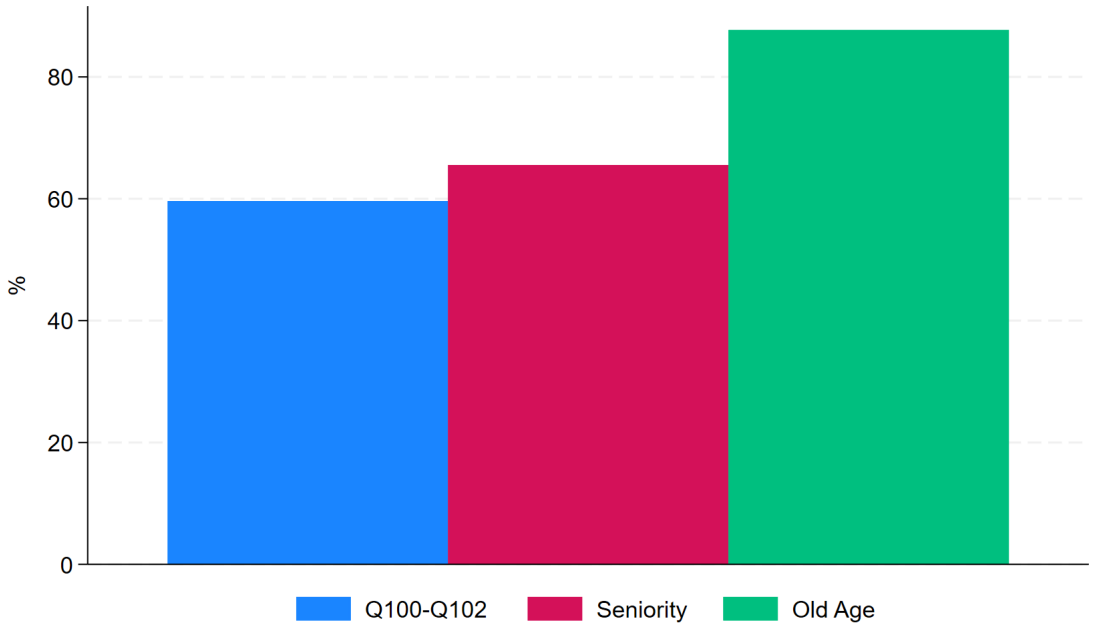
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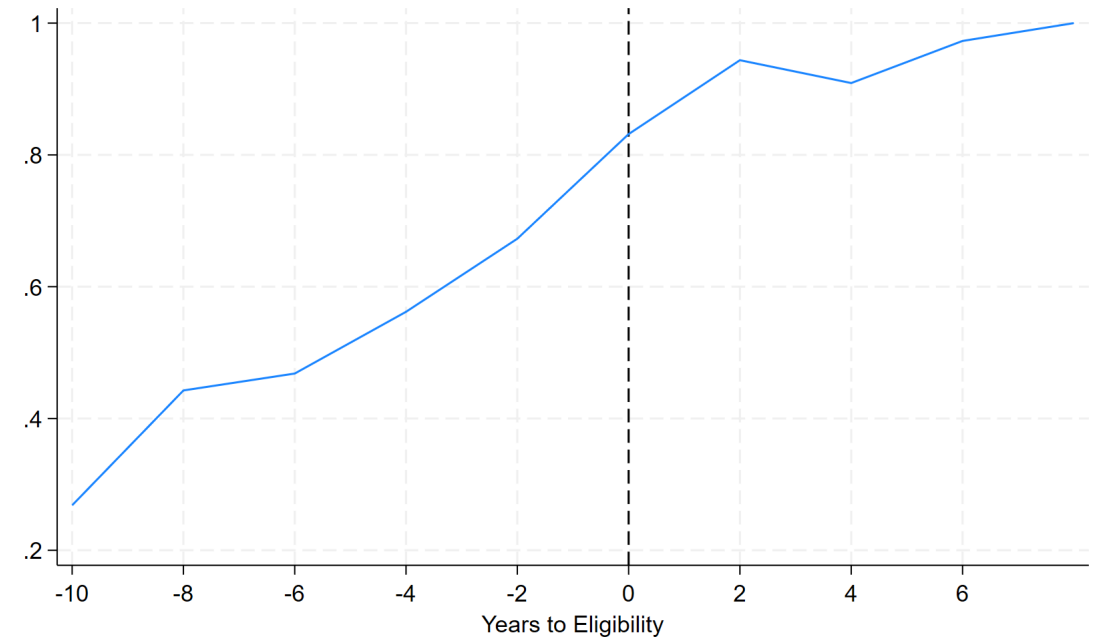
Figures

Figure 1: Retirement Options Take-Up



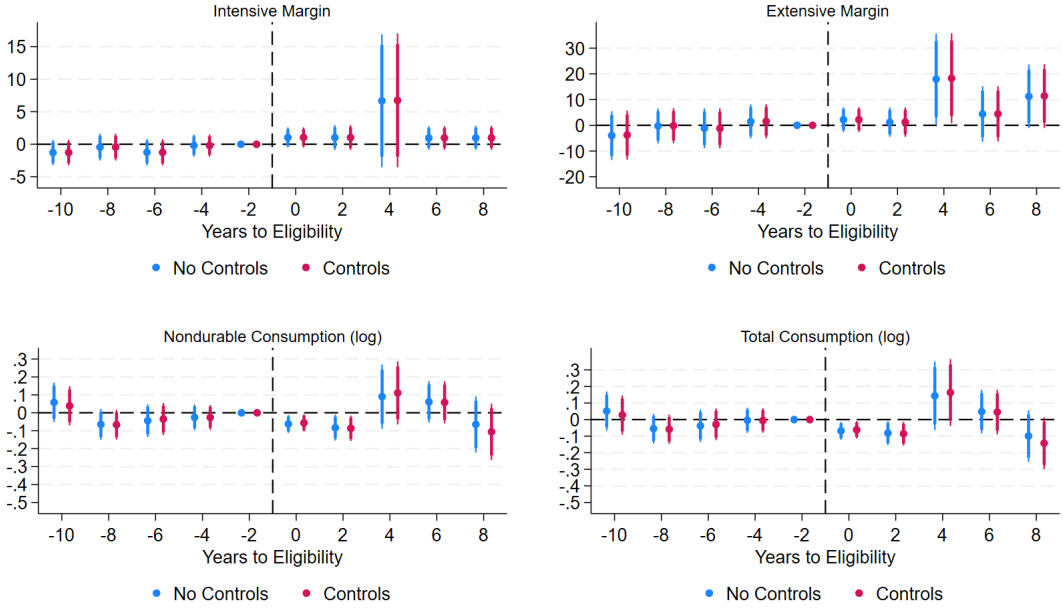
Notes: This Figure displays the take-up rates for the three retirement pathways: early retirement (blue bars), seniority-based retirement (red bars), and old-age retirement (green bars). Early retirement was available only between 2019 and 2022, corresponding to the Quota 100 and Quota 102 schemes (Q100-Q102).

Figure 2: Actual Retirement vs Eligibility



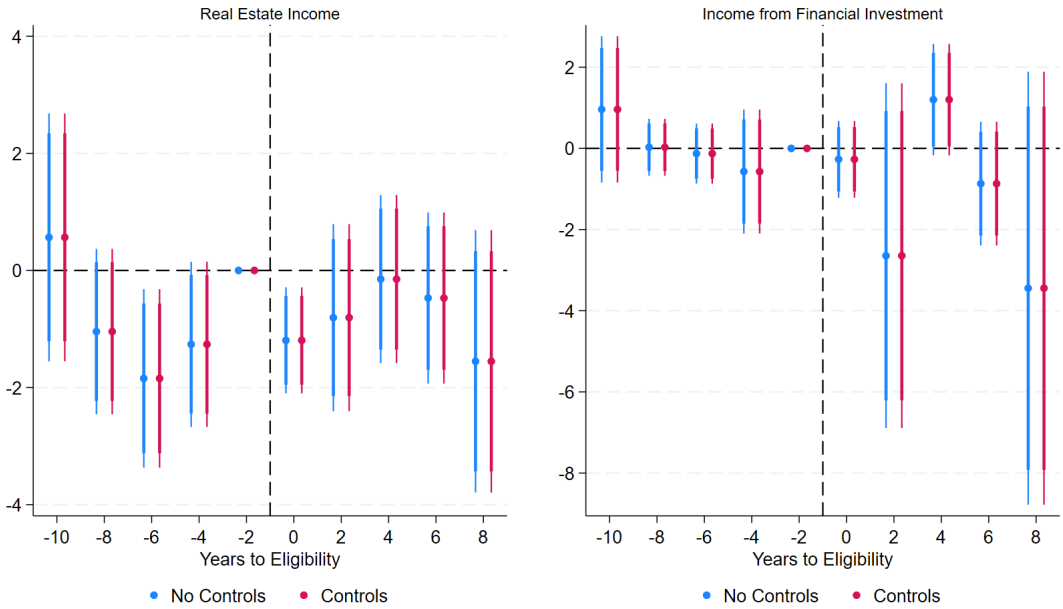
Notes: This Figure plots the relationship between actual retirement and the distance to eligibility. The vertical dashed line indicates the retirement threshold (distance = 0).

Figure 3: Pre-Trends in Stock Market Outcomes



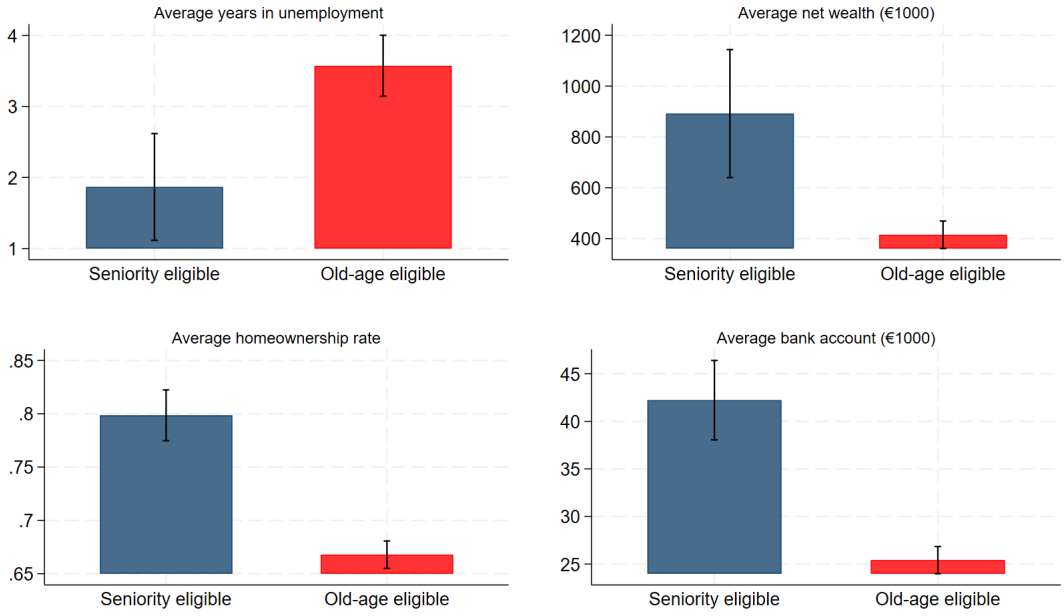
Notes: This figure shows the OLS estimated coefficients of equation (4). The left panel plots the evolution of the share of financial wealth held in stocks, while the right panel shows the probability of participating in the stock market. Each point represents the average difference relative to the year of eligibility (year 0), controlling for individual and time fixed effects. The vertical lines represent 95% confidence intervals.

Figure 4: Pre-trends in Real Estate and Financial Income



Notes: This figure shows the OLS estimated coefficients of equation (4). The left panel plots the evolution of the real estate income, while the right panel shows the income from financial investment. Each point represents the average difference relative to the year of eligibility (year 0), controlling for individual and time fixed effects. The vertical lines represent 95% confidence intervals.

Figure 5: Comparison of Old Age and Seniority Retirees



Notes: This Figure compares old-age retirees with seniority retirees on years spent in unemployment (top-left), net wealth (top-right), home ownership (bottom-left), and bank account balance (bottom-right). The vertical lines represent 95% confidence intervals.

Tables

Table 1: Summary Statistics from SHIW data by Eligibility Group

	Not Eligible		Early		Seniority		Old Age	
	Mean	SD	Mean	SD	Mean	SD	Mean	SD
Age	60.19	4.11	64.30	1.28	67.25	4.55	70.22	3.12
Married	0.68	0.47	0.78	0.42	0.73	0.45	0.65	0.48
Family Size	2.56	1.19	2.43	1.00	2.15	0.90	1.99	0.86
Centre	0.22	0.41	0.17	0.38	0.17	0.38	0.21	0.41
North	0.42	0.49	0.48	0.50	0.59	0.49	0.45	0.50
Medium Education	0.68	0.46	0.71	0.45	0.65	0.48	0.58	0.49
High Education	0.20	0.40	0.26	0.44	0.24	0.43	0.18	0.38
Years of Contribution	29.88	10.22	40.24	1.44	44.73	2.93	35.72	5.36
Retired	0.29	0.45	0.59	0.49	0.67	0.47	0.93	0.26
Nondurable Consumption	28.67	22.24	35.75	34.94	34.89	28.52	25.95	16.47
Cash on Hand (€1000)	175.20	2019.85	376.73	1531.40	255.06	926.94	112.86	524.04
Experienced long unemployment	0.19	0.39	0.07	0.26	0.04	0.19	0.09	0.28
Net Wealth (€1000)	563.28	5081.27	912.42	3077.71	891.83	4240.01	415.16	1979.24
Financial Assets (€1000)	58.94	156.42	139.37	272.87	117.11	226.96	57.43	140.18
Share of Stocks (x 100)	1.54	7.65	3.12	11.05	2.37	9.11	1.55	8.14
Participate in the Stock Market	7.97	27.08	16.74	37.37	13.74	34.44	7.12	25.73

Notes. This table reports descriptive statistics for individuals classified by their eligibility for different retirement schemes: not eligible, early retirement, seniority, and old age. Variables include different socio-economic and demographic characteristics, labor market experience, and financial indicators.

Table 2: The Effect of Retirement on Stock Holdings - All stocks

	Intensive Margin			Extensive Margin		
	(1)	(2)	(3)	(4)	(5)	(6)
Retired	7.46**	7.56**	7.57**	8.86	8.94	8.96
	(3.05)	(3.09)	(3.09)	(8.06)	(8.14)	(8.15)
Family Size		0.25	0.28		0.15	0.19
		(0.36)	(0.36)		(1.13)	(1.13)
Married		0.02	0.00		-1.39	-1.42
		(1.23)	(1.23)		(2.15)	(2.15)
Medium Education		-0.06	-0.06		0.05	0.05
		(0.28)	(0.28)		(0.46)	(0.46)
High Education		0.54	0.54		-2.39	-2.38
		(1.59)	(1.59)		(8.64)	(8.64)
Age		-0.16	-0.16		-0.14	-0.14
		(0.17)	(0.17)		(0.28)	(0.28)
Net Wealth (€1000)			0.00			0.00
			(0.00)			(0.00)
Observations	4676	4676	4676	5115	5115	5115
F-stat 1st stage	64.36	63.54	63.47	71.85	71.20	71.13

Notes: This Table shows the 2SLS estimates of equation (3). In columns 1 to 3, the dependent variable is the ratio of all stocks held by household i at time t to financial assets (intensive margin). From columns 4 to 6, the dependent variable is a dummy for stock market participation (extensive margin). *Retired* equals 1 if i is retired 0 otherwise, and is instrumented by *Eligible*, defined as in equation (2). Standard errors are clustered at the household level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 3: The Effect of Retirement on Stock Holdings - Public stocks

	Intensive Margin			Extensive Margin		
	(1)	(2)	(3)	(4)	(5)	(6)
Retired	7.37**	7.47**	7.48**	10.43	10.67	10.69
	(3.04)	(3.08)	(3.09)	(7.93)	(8.01)	(8.01)
Family Size		0.29	0.31		0.95	0.98
		(0.35)	(0.35)		(1.03)	(1.04)
Married		-0.17	-0.18		-1.90	-1.92
		(1.32)	(1.33)		(2.13)	(2.13)
Medium Education		-0.06	-0.06		0.01	0.01
		(0.28)	(0.28)		(0.51)	(0.51)
High Education		0.54	0.54		-2.21	-2.20
		(1.58)	(1.58)		(8.67)	(8.67)
Age		-0.16	-0.16		-0.22	-0.22
		(0.17)	(0.17)		(0.28)	(0.28)
Net Wealth (€1000)			0.00			0.00
			(0.00)			(0.00)
Observations	4676	4676	4676	5115	5115	5115

Notes: This Table shows the 2SLS estimates of equation (3). In columns 1 to 3, the dependent variable is the ratio of publicly traded stocks held by household i at time t to financial assets (intensive margin). From columns 4 to 6, the dependent variable is a dummy for stock market participation (extensive margin). *Retired* equals 1 if i is retired 0 otherwise, and is instrumented by *Eligible*, defined as in (2). Standard errors are clustered at the household level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 4: The Effect of Retirement on Stock Holdings - Private stocks

	Intensive Margin			Extensive Margin		
	(1)	(2)	(3)	(4)	(5)	(6)
Retired	0.09	0.09	0.09	-5.91*	-6.06*	-5.97*
	(0.17)	(0.17)	(0.17)	(3.31)	(3.35)	(3.34)
Family Size		-0.03	-0.03		-0.91	-0.72
		(0.03)	(0.03)		(0.68)	(0.63)
Married		0.19	0.19		2.39	2.25
		(0.17)	(0.17)		(2.06)	(2.04)
Medium Education		0.00	0.00		-0.02	-0.01
		(0.01)	(0.01)		(0.27)	(0.26)
High Education		-0.01	-0.01		-0.76	-0.71
		(0.02)	(0.02)		(0.59)	(0.57)
Age		-0.01	-0.01		0.08	0.07
		(0.01)	(0.01)		(0.14)	(0.14)
Net Wealth (€1000)			0.00			0.00
			(0.00)			(0.00)
Observations	4676	4676	4676	5115	5115	5115

Notes: This Table shows the 2SLS estimates of equation (3). In columns 1 to 3, the dependent variable is the ratio of private stocks held by household i at time t to financial assets (intensive margin). From columns 4 to 6, the dependent variable is a dummy for stock market participation (extensive margin). *Retired* equals 1 if i is retired 0 otherwise, and is instrumented by *Eligible*, defined as in (2). Standard errors are clustered at the household level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 5: Results on Consumption

	Nondurable Consumption (log)			Total Consumption (log)		
	(1)	(2)	(3)	(4)	(5)	(6)
Retired	-0.25**	-0.23**	-0.23**	-0.32***	-0.30**	-0.30**
	(0.12)	(0.11)	(0.11)	(0.12)	(0.12)	(0.12)
Observations	5115	5115	5115	5114	5114	5114

Notes: This Table shows the 2SLS estimates of equation (4). From columns 1 to 3, the dependent variable is either nondurable or total consumption by household i at time t over financial assets (intensive margin). From columns 4 to 6, the dependent variable is a dummy for stock market participation (extensive margin). *Retired* equals 1 if i is retired 0 otherwise, and is instrumented by *Eligible*, defined as in (2). Standard errors are clustered at the household level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 6: First Stage Estimation

	(1)	(2)	(3)
Eligible for Retirement	0.19***	0.18***	0.18***
	(0.02)	(0.02)	(0.02)
Family Size		-0.04***	-0.04***
		(0.01)	(0.01)
Married		-0.00	-0.00
		(0.05)	(0.05)
Medium Education		0.02	0.02
		(0.04)	(0.04)
High Education		-0.09	-0.09
		(0.07)	(0.07)
Age		0.01	0.01
		(0.02)	(0.02)
age_*			
Over 60			
Net Wealth (€1000)			-0.00***
			(0.00)
Observations	5115	5115	5115

Notes. This table reports the first-stage estimates from equation (4), where actual retirement status is regressed on retirement eligibility. Column (1) includes only individual and time fixed effects. Column (2) adds demographic controls. Column (3) further includes net wealth. Standard errors are clustered at the individual level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 7: Reduced Form Analysis

	Stock holdings		Consumption (log)	
	Intensive	Extensive	Nondurable	Total
	(1)	(2)	(3)	(4)
Seniority Eligible	2.36***	5.59**	-0.01	-0.05
	(0.86)	(2.72)	(0.03)	(0.03)
Q100 Eligible	1.14	1.06	-0.01	-0.01
	(0.88)	(2.88)	(0.04)	(0.04)
Old Age Eligible	1.12*	0.79	-0.06***	-0.07***
	(0.61)	(1.64)	(0.02)	(0.02)
Observations	4676	5115	5115	5114
Controls	✓	✓	✓	✓

Notes. This table reports the OLS estimates from equation (7). Standard errors are clustered at the individual level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

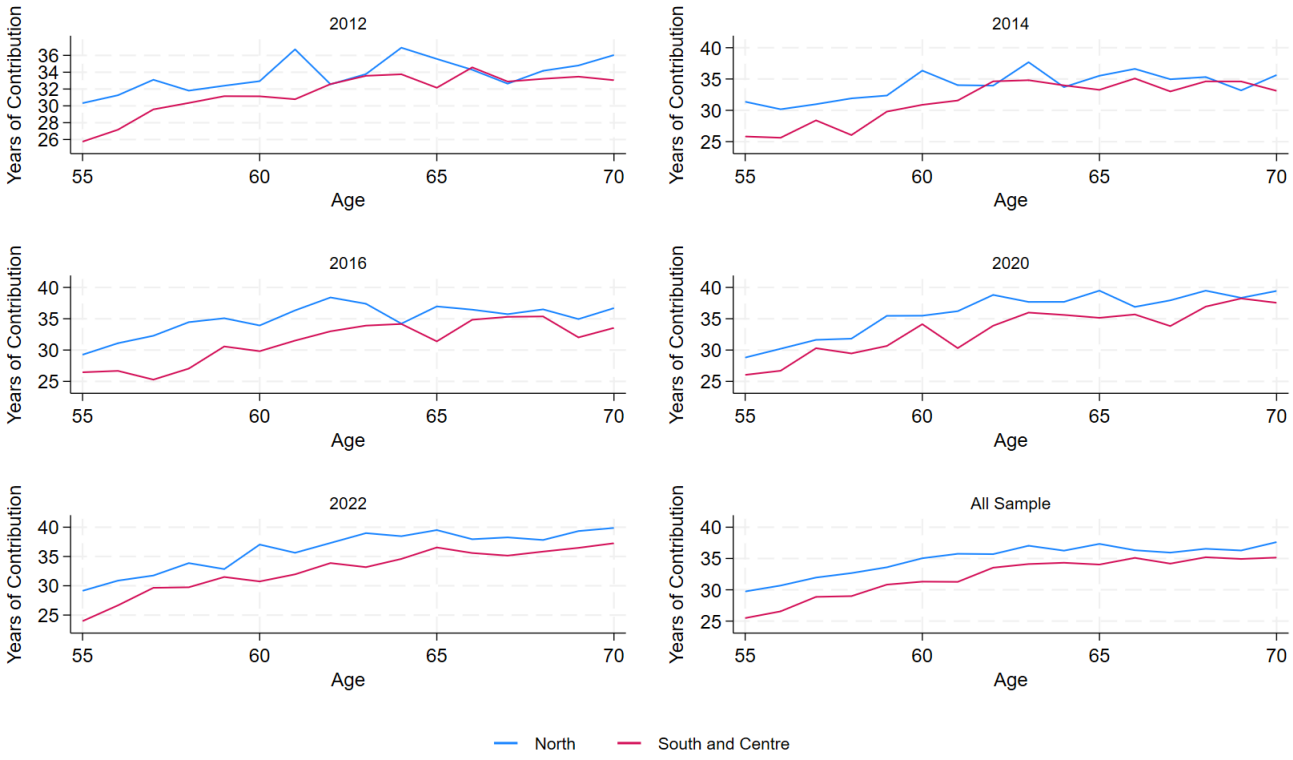
Table 8: Other Financial Assets

	Bank Account		Corporate Bonds		Government Bonds	
	(1)	(2)	(3)	(4)	(5)	(6)
Retired	-11.62	-2.53	5.21	5.44	5.74	9.52
	(11.33)	(9.16)	(6.57)	(11.70)	(6.63)	(10.23)
Observations	4676	5115	4676	5115	4676	5115

Notes. This table reports the 2SLS estimates from equation (3). The table title indicates the dependent variables. Odd numbers are the intensive margin, while even numbers are the extensive margin. Standard errors are clustered at the individual level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

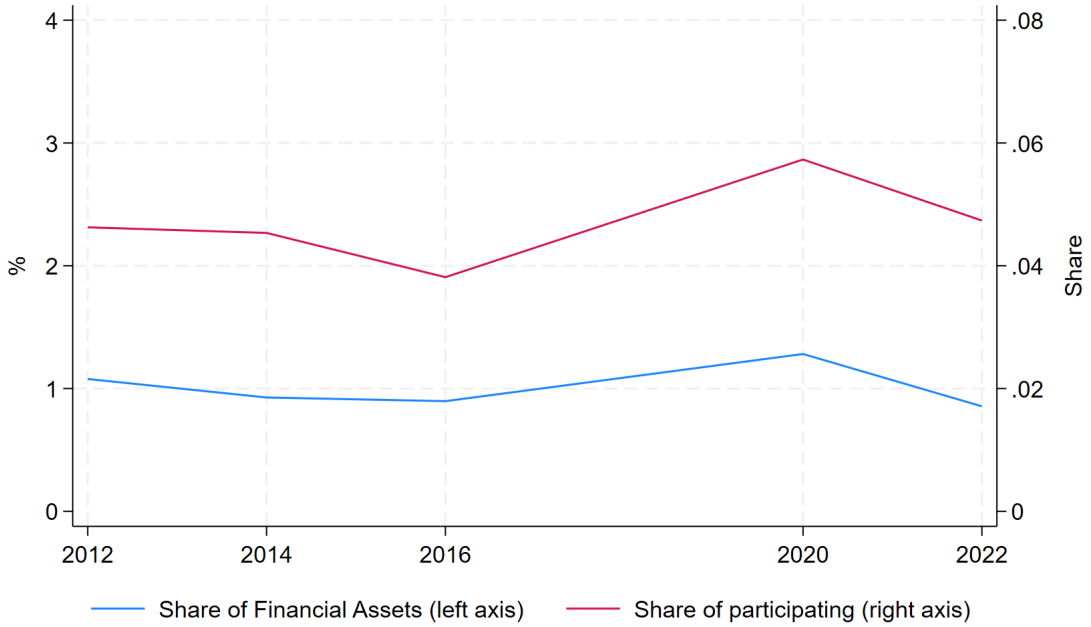
Appendix A: Additional Descriptive Statistics

Figure A1: Years of Contribution in 2020 and 2022 by Region



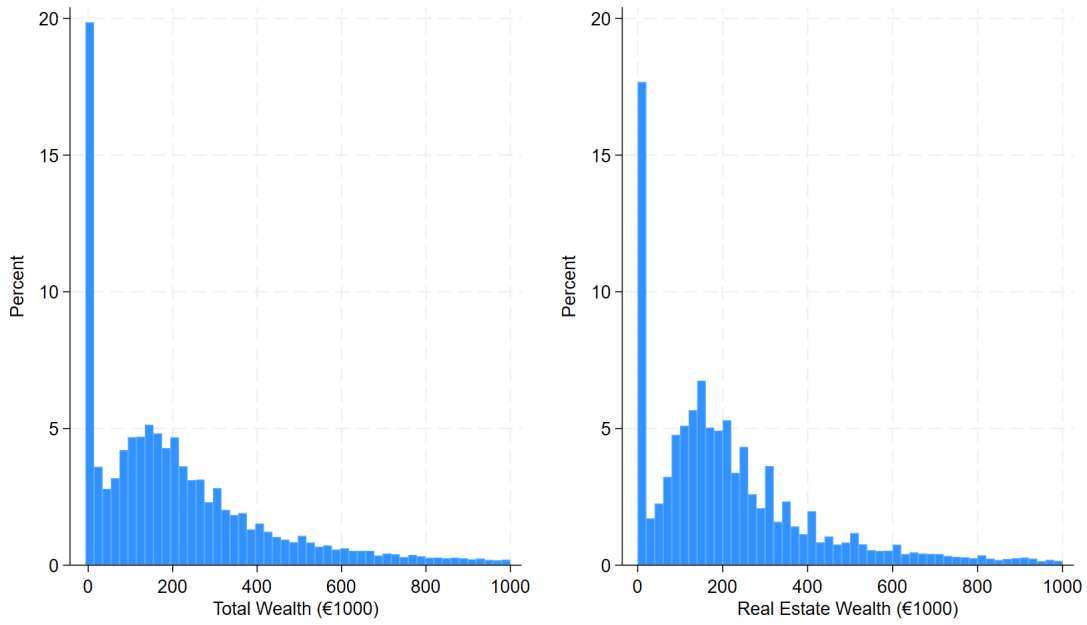
Notes: This Figure shows the average years of contribution by age, separately for individuals in the North (blue line) and in the South/Centre (red line) of Italy. The left panel shows 2020, and the right panel shows 2022. The vertical dashed line marks the minimum age requirement for early retirement (62 in 2020, 64 in 2022), while the horizontal dashed line indicates the contribution requirement (38 years).

Figure A2: Stock Market participation over time



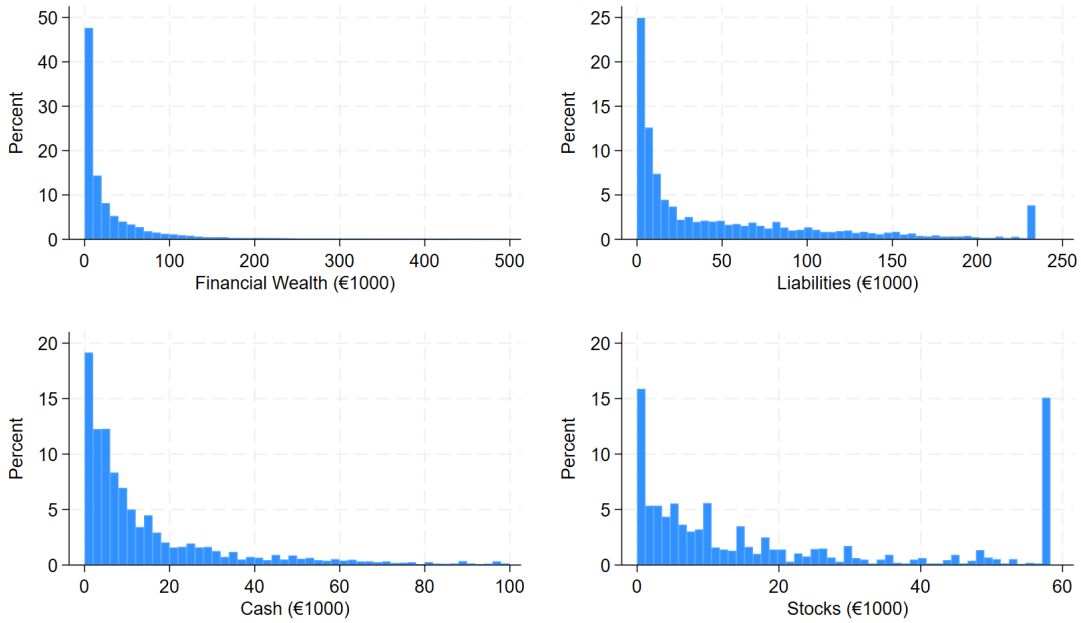
Notes: This Figure shows the average share of stock holdings of total financial assets between 2012 and 2022. The blue line (left axis) shows the percentage of stocks held as a share of financial assets; the red line (right axis) shows the fraction of people participating in the stock market. The year 2018 is missing.

Figure A3: Net Wealth and Real Estate Wealth



Notes: This Figure shows the pooled distribution of net wealth (left panel) and real estate wealth. Net wealth is the sum of financial wealth, real estate wealth, and financial liabilities.

Figure A4: Financial Wealth



Notes: The upper-left panel shows the pooled distribution of financial assets; the upper-right panel shows the distribution of financial liabilities. The bottom-left panel shows the distribution of cash, while the bottom-right panel shows the distribution of stocks.

Figure A5: Life Cycle Profiles of Consumption and Income



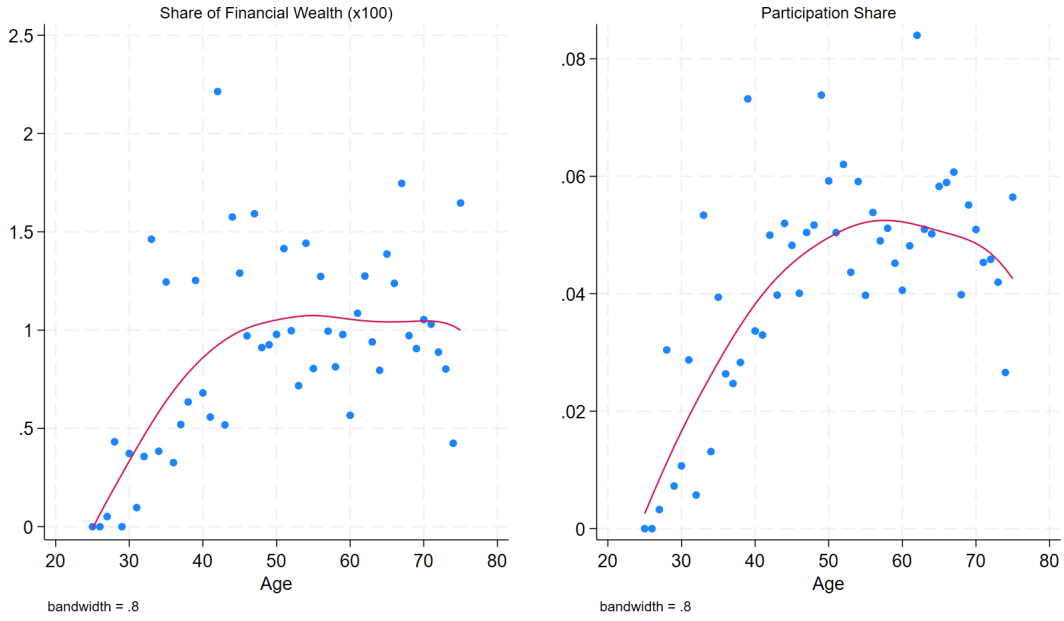
Notes: This Figure shows the life cycle profile of nondurable consumption (left panel) and of working income (right panel)

Figure A6: Life Cycle Profiles of Financial Assets and Stock Holding



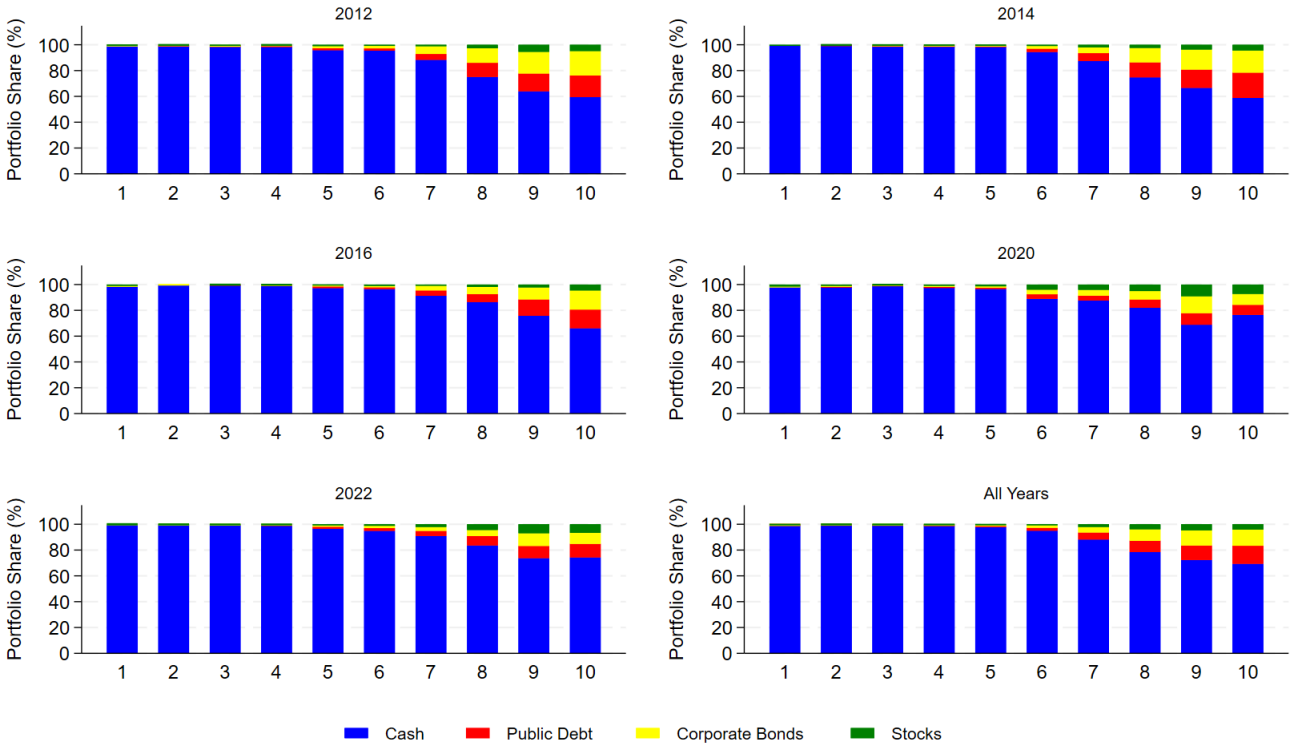
Notes: This Figure shows the life cycle profile of the value of financial assets (left panel) and the value of stocks (right panel)

Figure A7: Life Cycle Profiles of Stock Holding - Shares



Notes: This Figure shows the life cycle profile of the share of stocks over financial assets (left panel) and the stock market participation (right panel)

Figure A8: Composition of the Financial Portfolio



Notes: This Figure shows the average composition of financial wealth in all years, divided into cash (blue), government securities (red), corporate bonds (yellow), and stocks/mutual funds (green)

Appendix B: SHIW Data and Dataset Construction

The Survey of Household Income and Wealth (SHIW) started in the 1960s and collects information on the saving and investment behavior of Italian Households. In recent years, SHIW has asked retirement-specific questions, such as years of contribution, expected retirement age, and expected pension benefit as a share of last working earnings. In particular, we use the first piece of information, together with age, to define eligibility for the three retirement options explained in the main text.

We use the 2012, 2014, 2016, 2020, and 2022 waves and track the same household head. This sample selectiis follows the same criterion as **???**. Since we have no individual identifier, only a family identifier, we focus on the household head and ensure she does not change from one wave to another. We therefore drop households whose head changes sex, reports inconsistent ages, or shows inconsistent years of contribution over time. For instance, if i is 30 in 2012 and has worked for 5 years, she must be 32 and have worked for at most 7 years in 2014, 34 in 2014, and so on. This procedure yields a sample of around 2000 household-year observations.

SHIW has two types of datasets: historical and yearly. The historical dataset is a single dataset that includes all waves and contains variables aggregated at the household level. The yearly datasets contain more detailed information, such as years of contribution, in-depth demographics, wealth,h and employment status. We use both the historical and the yearly datasets, merging them for every year using the family unique identifier *nquest*. The table below describes the variables we use and how we derive them.

Table B1: List of Used Variables

Variable	**Units**	**Description and Source Variables**
nquest	number	cross-sectional identifier
year	year	time
age	number of years	
yc	number of years	years o contribution: acontrib
disann	number of years	year of unemployment
dislav	dummy	experienced long term unemployment
feale	dummy	gender: sesso == 2
low_ed	dummy	low education: studio $<$ 3
mid_ed	dummy	middle education (high school): studio == 4 or 5
high_ed	dummy	high education (college decree or more): studio $>$ 5
south	dummy	living in the South: area5 == 4 or 5
centre	dummy	living in the Centre: area5 == 3
north	dummy	living in the North: area5 == 1 or 2
family_size	number	household components: ncomp
retired	dummy	retired: qual $>$ 7
cn	€1000	nondurable consumption
cd	€1000	durable consumption
c	€1000	total consumption: cn + cd
lc	log	log of total consumption: $\log(c)$
lcn	log	log of nondurable consumption: $\log(cn)$

Continued on next page

Variable	Units	Description and Source Var ables
ytp	€1000	family pension earnings
yca	€1000	real estate income
ycf	€1000	financial income
ar	€1000	real estate wealth
af	€1000	total financial wealth
pf	€1000	total liabilities
w	€1000	net wealth: ar + af - pf
$bank_account$	€1000	amount of cash: ld
$public_debt$	€1000	government securities, af2
$bonds_private$	€1000	private bonds (companies and banks): lobb
$assvita$	€1000	amount paid annually for life insurance
$pensint$	€1000	amount paid annually for private pensions
$stocks_listed$	€1000	stocks traded on the stock market (market value at year-end): lazq
$stocks_unlisted$	€1000	stocks of private businesses: lazi + lsrl + lper - stocks listed until 2016; lazi2 - stocks listed from 2020
$stocks$	€1000	total stocks: $stocks_listed + stocks_unlisted$
$managed_savings$	€1000	managed savings: lgp
$others$	€1000	other assets: lat
$foreign$	€1000	foreign assets: lte
$share_bank_account$	%	share of cash: bank account / af
$share_public_debt$	%	share of government securities, public debt / af
$share_bonds_private$	%	share of private bonds: bonds private / af
$share_stocks_listed$	%	share of listed stocks: stocks listed / af
$share_stocks_unlisted$	%	share of stocks of private businesses: stocks unlisted / af
$share_stocks$	%	share of total stocks: stocks / af
$share_managed$	%	share of managed savings: managed / af
$share_others$	%	share of other assets: others/ af
$share_foreign$	%	share of foreign assets: foreign / af
$p_bank_account$	dummy	hods cash: $bank_account > 0$
p_public_debt	dummy	holds government securities: $public_debt > 0$
$p_bonds_private$	dummy	holds private bonds: $bonds_private > 0$
p_stocks_listed	dummy	holds listed stocks: $stocks_listed > 0$
$p_share_stocks_unlisted$	dummy	holds stocks of private businesses: $stocks_unlisted > 0$
p_share_stocks	dummy	holds total stocks: $stocks > 0$
p_share_others	dummy	holds other assets: $others > 0$
$p_share_foreign$	dummy	holds foreign assets: $foreign > 0$